

Milestone #: 2

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Group Number: 63

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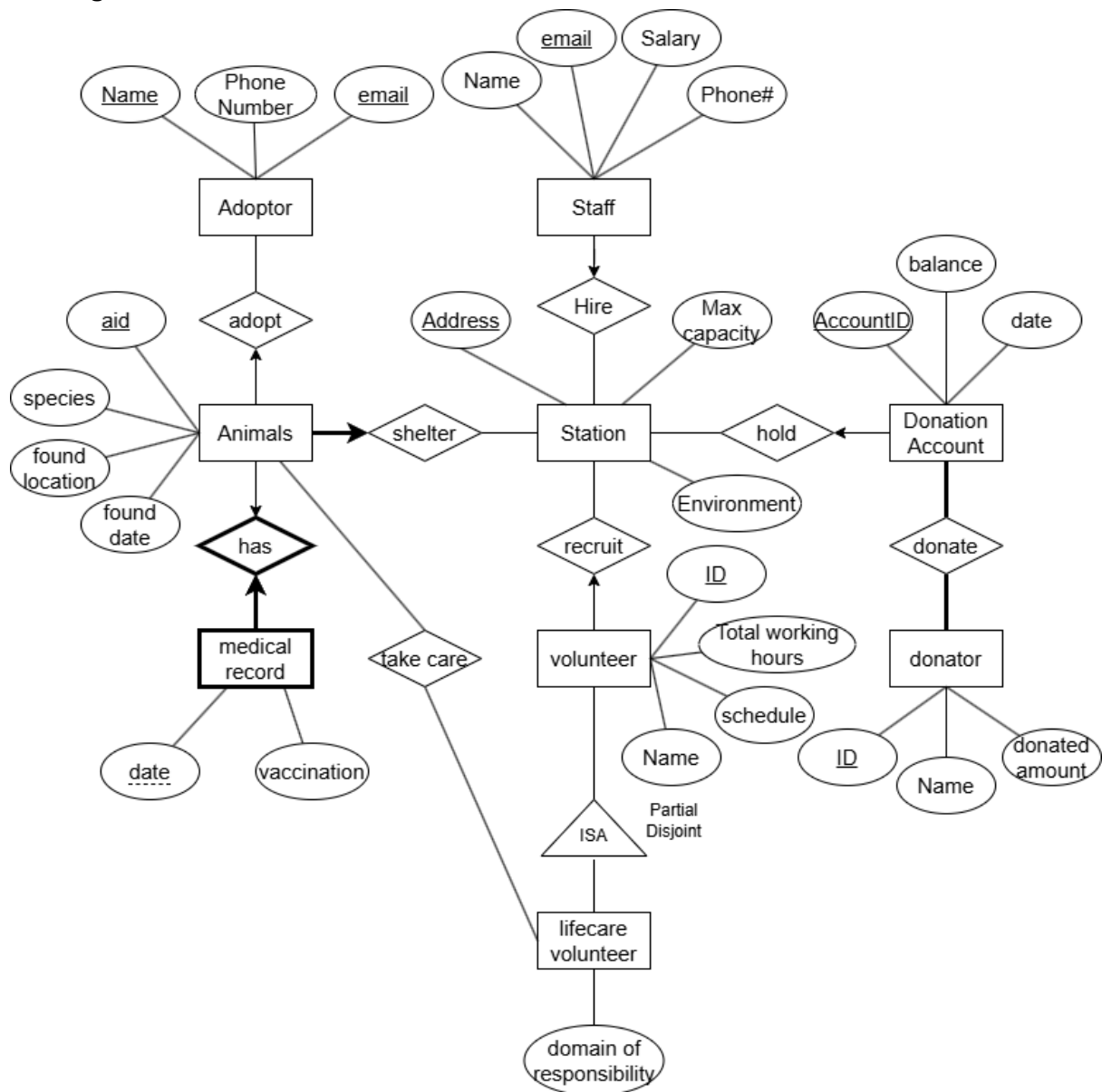
By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Brief summary:

Our project focuses on the database for management for rescued animals. It tracks animal details (species, rescue history), volunteer activities (schedules, specialized roles), adoptors, and medical care (vaccinations), while coordinating shelter resources (station capacity, donation funds).

ER diagram :



Change:

We modified the attributes in adaptor and staff. We changed the key to "email" in both entities. This can avoid having to use an artificial id.

Schema:

MedicalRecord_Has(recordDate,animalID,vaccination)

Adaptor(email,name,phone number)

Animals_Shelter_Adopt(animalID, email,Address ,species, found location, found date)

Volunteer_recruit(address,VolunteerID, total working hours,name,schedule)

Station(address,max capacity,environment)
Staff_Hire(email,salary,phone number,name,*address*)
lifecare volunteer(ID,domain of responsibility)
Donator(DID, name,donated account)
Donation Account_hold(accountID, *address*,balance,date)
shelter(animalid,*address*)
takecare(animalID,VolunteerID)

FDs in normal form BCNFs:

MedicalRecord_Has(recordDate,animalID,vaccination):
recordDate,*animalID* → vaccination

Animals_shelter_Adopt(animalID, *email*,*Address* ,species, found location, found date)
animalID → species, found location, found date,*Address* ,*email*

Staff_Hire(email,salary,phone number,name,*address*):
email → name,*address*, salary,phone_number

Donation Account_hold(accountID, *address*,balance,date)
accountID → *address*,balance,date

takecare (animalID, VolunteerID)

Volunteer_recruit(*address*,VolunteerID)
VolunteerID → *address*

SQL code (including DDL and insert operations)

```
DROP TABLE TakeCare CASCADE CONSTRAINTS;
DROP TABLE MedicalRecord_Has CASCADE CONSTRAINTS;
DROP TABLE Donation_Account_Hold CASCADE CONSTRAINTS;
DROP TABLE Donator CASCADE CONSTRAINTS;
DROP TABLE Lifecare_Volunteer CASCADE CONSTRAINTS;
DROP TABLE Volunteer_Recruit CASCADE CONSTRAINTS;
DROP TABLE Staff_Hire CASCADE CONSTRAINTS;
DROP TABLE Animals_Adopt_Shelter CASCADE CONSTRAINTS;
DROP TABLE Station CASCADE CONSTRAINTS;
DROP TABLE Adopter CASCADE CONSTRAINTS;

CREATE TABLE Adopter (
    email          CHAR(50) ,
    name           CHAR(50) NOT NULL,
```

```

    phone_number CHAR(20) UNIQUE NOT NULL,
    PRIMARY KEY(email)
);

CREATE TABLE Station (
    address      CHAR(255) PRIMARY KEY,
    max_capacity INT NOT NULL,
    environment  CHAR(255) NOT NULL
);

CREATE TABLE Animals_Adopt_Shelter (
    aid          INT PRIMARY KEY,
    species      CHAR(40) NOT NULL,
    found_location CHAR(50) NOT NULL,
    found_date   DATE NOT NULL,
    email        CHAR(50),
    address      CHAR(255),
    FOREIGN KEY (email) REFERENCES Adopter(email) ON DELETE
SET NULL,
    FOREIGN KEY (address) REFERENCES Station(address) ON DELETE
CASCADE
);

CREATE TABLE MedicalRecord_Has (
    recordDate  DATE,
    aid         INT PRIMARY KEY,
    vaccination CHAR(1),
    FOREIGN KEY (aid) REFERENCES Animals_Adopt_Shelter(aid) ON
DELETE CASCADE
);

CREATE TABLE Staff_Hire (
    email      CHAR(50),
    salary     DECIMAL(10,2) NOT NULL,

```

```
phone_number CHAR(20) UNIQUE NOT NULL,  
name          CHAR(50) NOT NULL,  
address       CHAR(255) NOT NULL,  
PRIMARY KEY(email),  
FOREIGN KEY (address) REFERENCES Station(address) ON DELETE  
SET NULL  
);
```

```
CREATE TABLE Volunteer_Recruit (  
    ID          INT,  
    total_working_hours INT NOT NULL,  
    name        CHAR(50) NOT NULL,  
    schedule    INT,  
    address     CHAR(255) NOT NULL,  
    PRIMARY KEY(ID),  
    FOREIGN KEY (address) REFERENCES Station(address) ON DELETE  
CASCADE  
);
```

```
CREATE TABLE Lifecare_Volunteer (  
    ID          INT,  
    domain_of_responsibility CHAR(50) NOT NULL,  
    PRIMARY KEY(ID),  
    FOREIGN KEY (ID) REFERENCES Volunteer_Recruit(ID) ON DELETE  
CASCADE  
);
```

```
CREATE TABLE Donator (  
    DID INT,  
    name CHAR(50) NOT NULL,  
    PRIMARY KEY(DID)  
);
```

```
CREATE TABLE Donation_Account_Hold (  

```

```

    accountID      INT PRIMARY KEY,
    balance         INT NOT NULL,
    donation_date  DATE NOT NULL,
    address         CHAR(255),
    FOREIGN KEY (address) REFERENCES Station(address) ON DELETE
CASCADE
);

```

```

CREATE TABLE TakeCare (
    aid INT NOT NULL,
    ID  INT NOT NULL,
    PRIMARY KEY (aid, ID),
    FOREIGN KEY (aid) REFERENCES Animals_Adopt_Shelter(aid) ON
DELETE CASCADE,
    FOREIGN KEY (ID)  REFERENCES Volunteer_Recruit(ID)      ON
DELETE CASCADE
);

```

-- 3.1 Adopter

```

INSERT INTO Adopter (email, name, phone_number)
VALUES ('abc@hotmail.com', 'p1', '123-456-7890');

```

```

INSERT INTO Adopter (email, name, phone_number)
VALUES ('bcd@gmail.com', 'p2', '234-567-1283');

```

```

INSERT INTO Adopter (email, name, phone_number)
VALUES ('cde@gmail.com', 'p3', '345-678-4761');

```

```

INSERT INTO Adopter (email, name, phone_number)
VALUES ('def@163.com', 'p4', '456-789-4756');

```

```

INSERT INTO Adopter (email, name, phone_number)
VALUES ('efg@gmail.com', 'p5', '125-732-5629');

```

```
-- 3.2 Station
INSERT INTO Station (address, max_capacity, environment)
VALUES ('Station A', 50, 'Urban');

INSERT INTO Station (address, max_capacity, environment)
VALUES ('Station B', 30, 'Suburban');

INSERT INTO Station (address, max_capacity, environment)
VALUES ('Station C', 20, 'Rural');

INSERT INTO Station (address, max_capacity, environment)
VALUES ('Station D', 100, 'Urban');

INSERT INTO Station (address, max_capacity, environment)
VALUES ('Station E', 40, 'Coastal');

-- 3.3 Staff_Hire
INSERT INTO Staff_Hire (email, salary, phone_number, name,
address)
VALUES ('micheal@rescue.com', 3000.00, '601-145-2345',
'Michael Williams', 'Station A');

INSERT INTO Staff_Hire (email, salary, phone_number, name,
address)
VALUES ('sarah@rescue.com', 3500.00, '779-311-2807', 'Sarah
Jones', 'Station B');

INSERT INTO Staff_Hire (email, salary, phone_number, name,
address)
VALUES ('david@rescue.com', 3200.00, '683-190-3598', 'David
Smith', 'Station A');
```

```
INSERT INTO Staff_Hire (email, salary, phone_number, name,
address)
VALUES ('emma@rescue.com', 2800.00, '106-326-1287', 'Emma
Rodriguez', 'Station C');

INSERT INTO Staff_Hire (email, salary, phone_number, name,
address)
VALUES ('james@rescue.com', 4000.00, '587-446-1677', 'James
Brown', 'Station E');

INSERT INTO Staff_Hire (email, salary, phone_number, name,
address)
VALUES ('lucy@rescue.com', 4500.00, '264-267-3640', 'Lucy
James', 'Station D');

-- 3.4 Volunteer_Recruit
INSERT INTO Volunteer_Recruit (ID, total_working_hours, name,
schedule, address)
VALUES (1, 20, 'Tom', 1, 'Station A');

INSERT INTO Volunteer_Recruit (ID, total_working_hours, name,
schedule, address)
VALUES (2, 30, 'Jerry', 2, 'Station A');

INSERT INTO Volunteer_Recruit (ID, total_working_hours, name,
schedule, address)
VALUES (3, 15, 'Maggie', 3, 'Station B');

INSERT INTO Volunteer_Recruit (ID, total_working_hours, name,
schedule, address)
VALUES (4, 25, 'Finn', 4, 'Station B');

INSERT INTO Volunteer_Recruit (ID, total_working_hours, name,
schedule, address)
```



```
VALUES (5, 10, 'Sophie', 5, 'Station C');

-- 3.5 Lifecare_Volunteer
INSERT INTO Lifecare_Volunteer (ID, domain_of_responsibility)
VALUES (1, 'Medical Care');

INSERT INTO Lifecare_Volunteer (ID, domain_of_responsibility)
VALUES (3, 'Nutrition');

INSERT INTO Lifecare_Volunteer (ID, domain_of_responsibility)
VALUES (5, 'Behavior Training');

-- 3.6 Donator
INSERT INTO Donator (DID, name)
VALUES (101, 'Charity Org');

INSERT INTO Donator (DID, name)
VALUES (102, 'John Donor');

INSERT INTO Donator (DID, name)
VALUES (103, 'Jane Donor');

INSERT INTO Donator (DID, name)
VALUES (104, 'ABC Foundation');

INSERT INTO Donator (DID, name)
VALUES (105, 'XYZ Philanthropy');

-- 3.7 Animals Adopt Shelter
INSERT INTO Animals_Adopt_Shelter
(aid, species, found_location, found_date, email, address)
VALUES (100, 'Dog', 'City Park', DATE '2023-01-10',
'abc@hotmail.com', 'Station A');
```

```
INSERT INTO Animals_Adopt_Shelter
(aid, species, found_location, found_date, email, address)
VALUES (101, 'Cat', 'Downtown', DATE '2023-01-20',
'bcd@gmail.com', 'Station B');

INSERT INTO Animals_Adopt_Shelter
(aid, species, found_location, found_date, email, address)
VALUES (102, 'Rabbit', 'Suburbs', DATE '2022-12-01',
'cde@gmail.com', 'Station C');

INSERT INTO Animals_Adopt_Shelter
(aid, species, found_location, found_date, email, address)
VALUES (103, 'Parrot', 'TropicalGarden', DATE '2022-11-15',
'def@163.com', 'Station D');

INSERT INTO Animals_Adopt_Shelter
(aid, species, found_location, found_date, email, address)
VALUES (104, 'Dog', 'School Yard', DATE '2023-02-10',
'efg@gmail.com', 'Station E');

-- 3.8 Donation_Account_Hold
-- 如果 address 外键要引用Station(address), 就用已有 'Station
A' ~ 'Station E'

INSERT INTO Donation_Account_Hold (accountID, balance,
donation_date, address)
VALUES (101, 5000, DATE '2024-01-10', 'Station A');

INSERT INTO Donation_Account_Hold (accountID, balance,
donation_date, address)
VALUES (102, 3000, DATE '2024-02-05', 'Station B');

INSERT INTO Donation_Account_Hold (accountID, balance,
donation_date, address)
VALUES (103, 7000, DATE '2024-02-20', 'Station C');
```

```
INSERT INTO Donation_Account_Hold (accountID, balance,
donation_date, address)
VALUES (104, 2000, DATE '2024-03-01', 'Station D');

INSERT INTO Donation_Account_Hold (accountID, balance,
donation_date, address)
VALUES (105, 4500, DATE '2024-03-15', 'Station E');

-- 3.9 TakeCare
--      aid=101,102,103,104,105 都必须已经在Animals_Adopt_Shelter里
插入
INSERT INTO TakeCare (aid, ID)
VALUES (101, 1);

INSERT INTO TakeCare (aid, ID)
VALUES (102, 2);

INSERT INTO TakeCare (aid, ID)
VALUES (103, 3);

INSERT INTO TakeCare (aid, ID)
VALUES (104, 1);

INSERT INTO TakeCare (aid, ID)
VALUES (100, 4);

-- 3.10 MedicalRecord_Has
--      Oracle里多行INSERT只能一条条写
INSERT INTO MedicalRecord_Has (recordDate, aid, vaccination)
VALUES (DATE '2024-02-01', 101, 'Y');

INSERT INTO MedicalRecord_Has (recordDate, aid, vaccination)
VALUES (DATE '2024-02-15', 102, 'N');
```

```
INSERT INTO MedicalRecord_Has (recordDate, aid, vaccination)
VALUES (DATE '2024-03-05', 103, 'Y');
```

```
INSERT INTO MedicalRecord_Has (recordDate, aid, vaccination)
VALUES (DATE '2024-03-10', 104, 'Y');
```

```
INSERT INTO MedicalRecord_Has (recordDate, aid, vaccination)
VALUES (DATE '2024-03-20', 100, 'N');
```

AI declaration:

We used chatGPT to generate some data in the insertion part.